BASE CHANGE COMPATIBILITY OF THE DERIVED PUSHFORWARD OF A TENSOR PRODUCT WITH A FLAT COMPLEX

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The base change isomorphism for a single complex holds exactly when the canonical map from the derived pullback to the ordinary pullback satisfies the three equivalent conditions, and these conditions are inherited by suitable maps. invoked at proof lines 6909-6916. Reduces the tensor base change assertion to checking one condition for the map L(g')^*G to (g')^*G.
- For a complex of quasi-coherent modules flat over the base, the derived tensor with the base change ring computes as the ordinary tensor on stalks. used at proof lines 6917-6925. Verifies the stalk condition for the canonical map using flatness of G over S.

Source reference. Original lemma 0DJ9: sources/stacks-perfect/pinned-original-source.tex, source lines 6749–6749. Cited at proof lines 6909-6912 as the first reduction input; the assembly of the reduction and the stalk check are inside the counted span.

Source reference. Original lemma 0DJA: sources/stacks-perfect/pinned-original-source.tex, source lines 6824–6824. Cited at proof lines 6909-6912 as the second reduction input; the application to the canonical map of the flat complex is inside the counted span.

Source reference. Original lemma 0DJ8: sources/stacks-perfect/pinned-original-source.tex, source lines 6654–6654. Cited at proof lines 6913-6916 as the target condition; its verification on stalks for the flat complex is inside the counted span.

Lemma 0.2. *Let $f : X \to S$ be a quasi-compact and quasi-separated morphism of schemes. Let $E \in D_{QCoh}(\mathcal{O}_X)$. Let $\mathcal{G}^\bullet$ be a bounded above complex of quasi-coherent $\mathcal{O}_X$-modules flat over S. Then formation of*

$$Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet)$$

commutes with arbitrary base change (see proof for precise statement).

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Proof. The statement means the following. Let $g : S' \to S$ be a morphism of schemes and consider the base change diagram

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

in other words $X' = S' \times_S X$. The lemma asserts that

$$Lg^* Rf_*(E \otimes^{\mathbf{L}}_{\mathcal{O}_X} \mathcal{G}^\bullet) \longrightarrow Rf'_* \left(L(g')^* E \otimes^{\mathbf{L}}_{\mathcal{O}_{X'}} (g')^* \mathcal{G}^\bullet \right)$$

is an isomorphism. Observe that on the right hand side we do **not** use the derived pullback on $\mathcal{G}^\bullet$. To prove this, we apply Lemmas 0DJ9 and 0DJA to see that it suffices to prove the canonical map

$$L(g')^* \mathcal{G}^\bullet \to (g')^* \mathcal{G}^\bullet$$

satisfies the equivalent conditions of Lemma 0DJ8. This follows by checking the condition on stalks, where it immediately follows from the fact that $\mathcal{G}^\bullet_x \otimes_{\mathcal{O}_{S,s}} \mathcal{O}_{S',s'}$ computes the derived tensor product by our assumptions on the complex $\mathcal{G}^\bullet$. $\square$